APPLIED MATH FOR COMPUTER SCIENCE
CPCS 212

LAB 1 : Introduction to MATLAB

1st Term 2024

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- Please write the subject of the email as fellow:

  CPCS212_ Section_Name_Subject

Lab Assessment

- Grading:

Weekly Activates and HWs	10
Final Lab Exam	10
Total Lab	20 marks

- Lab activities: to be uploaded at the end of each lab (diary file).
- Lab Homework: as practice for some labs

Agenda

- Introduction to the MATLAB
- Starting Up Matlab
- Desktop Environment
- using Matlab as a numerical calculator
- MATLAB Editor - Script files
- Getting Help
- Variable Names
    - Built-in Matlab Variable

Objectives

- After this lab, you should be:
- Able to start MATLAB
- Able to use MATLAB as calculator
- Able to use MATLAB Editor - Script files
- Able to use MATLAB help

Introduction

- The labs are conducted using "Matlab".
- Matlab is a program produced by The MathWorks.
- The Matlab language is an object-oriented programming language
- Matlab language has an extensive library of mathematical and scientific function calls entirely built-in.

What is MATLAB ?

- MATLAB (Matrix Algebra laboratory), distributed by The MathWorks, is a technical computing environment for high performance numeric computation and visualization.
- It integrates numerical analysis, matrix computation, signal processing, and graphics in an easy-to-use environment. MATLAB also features a family of application-specific solutions called toolboxes.
- Toolboxes are comprehensive collections of MATLAB functions that extend its environment in order to solve particular classes of problems

Using of MATLAB ?

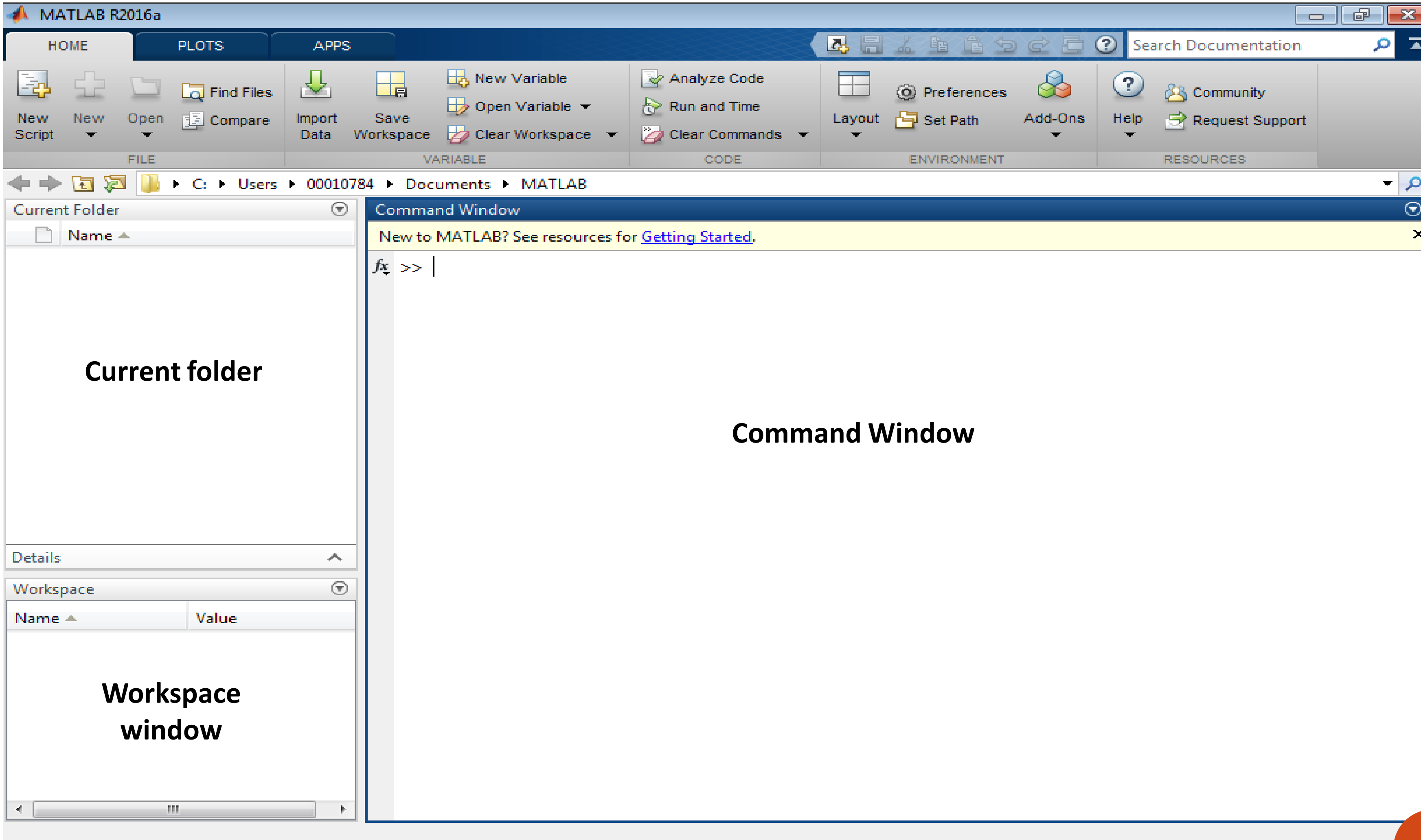
using Matlab as a numerical calculator

- entering row vectors and column vectors
- entering matrices
- forming matrix and vector products
- doing matrix products, sums etc
- using Matlab to solve linear equations
- Matlab functions that operate on arrays
- Plotting basic graphs using Matlab.

Starting Up Matlab

- There are two MATLAB releases each year
- This course is based on R2016a corresponds to MATLAB 9.0
- In the lab, you will find Matlab under **Start | All Programs | Matlab.**

How to get started ?

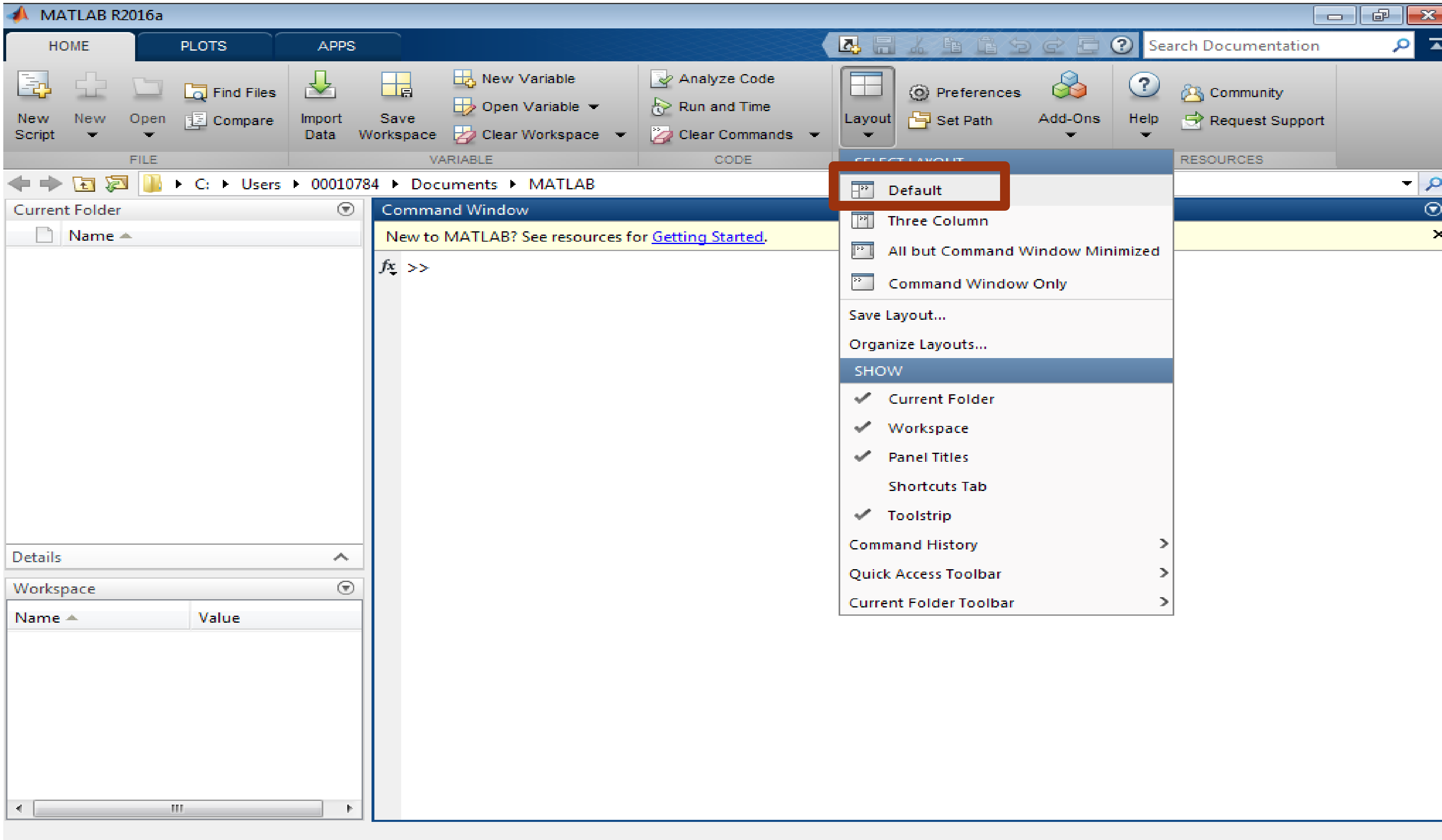


Desktop Environment

- MATLAB has its own Start menu
- MATLAB uses a three-column layout as the default:
  - **Command Window**
  - **Workspace**
  - **Current Folder**

If you close any of the above windows,
How can you restore the default MATLAB
Desktop again?

Desktop Environment



Window	Description
Command window	Used to type instructions of various types called commands, functions and statements.
Command History Window	•This window shows all the previous keystrokes you entered in the command window. •You can select a command from this window with the mouse and execute it in the command window by double clicking on it. •You can select a set of commands from this window and create an M-file with the right click of the mouse and selecting the appropriate option form the menu.
Current Directory Window	•Use it to access files •Double clicking on a file name with the extension ".m" will open that file in the MATLAB editor
Workspace window	•Displays the variables created in the command window •You can do various things with these variables, such as **Plotting, Edit value, Delete, Copy and Rename by clicking on a variable and then using the right button on the mouse to select your option.**

diary file

Matlab provides the capability to keep a record of *both* the commands *and* their output.

diary diary.txt

Required

- In first of each lab use

  diary yourID_labno.txt

- You should complete a report of the results you obtained for each completed lab.

Reports

The report consists of at least two files:

- The original, unchanged, diary.txt file(s)
- The summary file:

The diary.txt file and deleting all your false starts and errors

MATLAB comments:

- The comment symbol (%) may be put anywhere in the line. MATLAB ignores everything to the right of the % symbol.

```
>> % This is a comment line
```

Lab Exercise

- Organize your work into directories.

Create folder with name lab212

```
>> diary lab1diary.txt
```

- Try command "ver" which tells you which MATLAB features are installed.

```
>> ver
```

- Use the "path" command to see where MATLAB is looking for files.

```
>> path
```

Using Matlab as a numerical calculator

The Basic Arithmetic is:

- + Plus; addition operator.
- - Minus; subtraction operator.
- * Multiplication operator.
- / division operator.
- ^ Exponentiation (power) operator.

Lab Exercise

Expression	Example	Result	Comment
+	>> 5+14	ans= 19	Arithmetic works as expected. Note that the result is given the name "ans" each time.
-	>> 10-4	ans= 6	
*	>> 3*3	ans= 9	
/	>> 15/3	ans= 5	
^	>> 2^3	ans= 8	

MATLAB Editor (Script files)

You can perform operations in MATLAB in two ways:

- In the interactive mode *(similar to using a calculator)*, in which all commands are entered directly in the Command window,

OR

- By running a MATLAB program stored in *script* file. This type of file contains MATLAB commands, so running it is equivalent to typing all the commands - one at a time - at the Command window prompt.

MATLAB Editor (Script files)

- MATLAB Editor is a text editor with some special features to make it useful for working with MATLAB code.

- MATLAB Editor used to create the script file which allows you to write your own programs and save in M-files, which have the extension ".m"

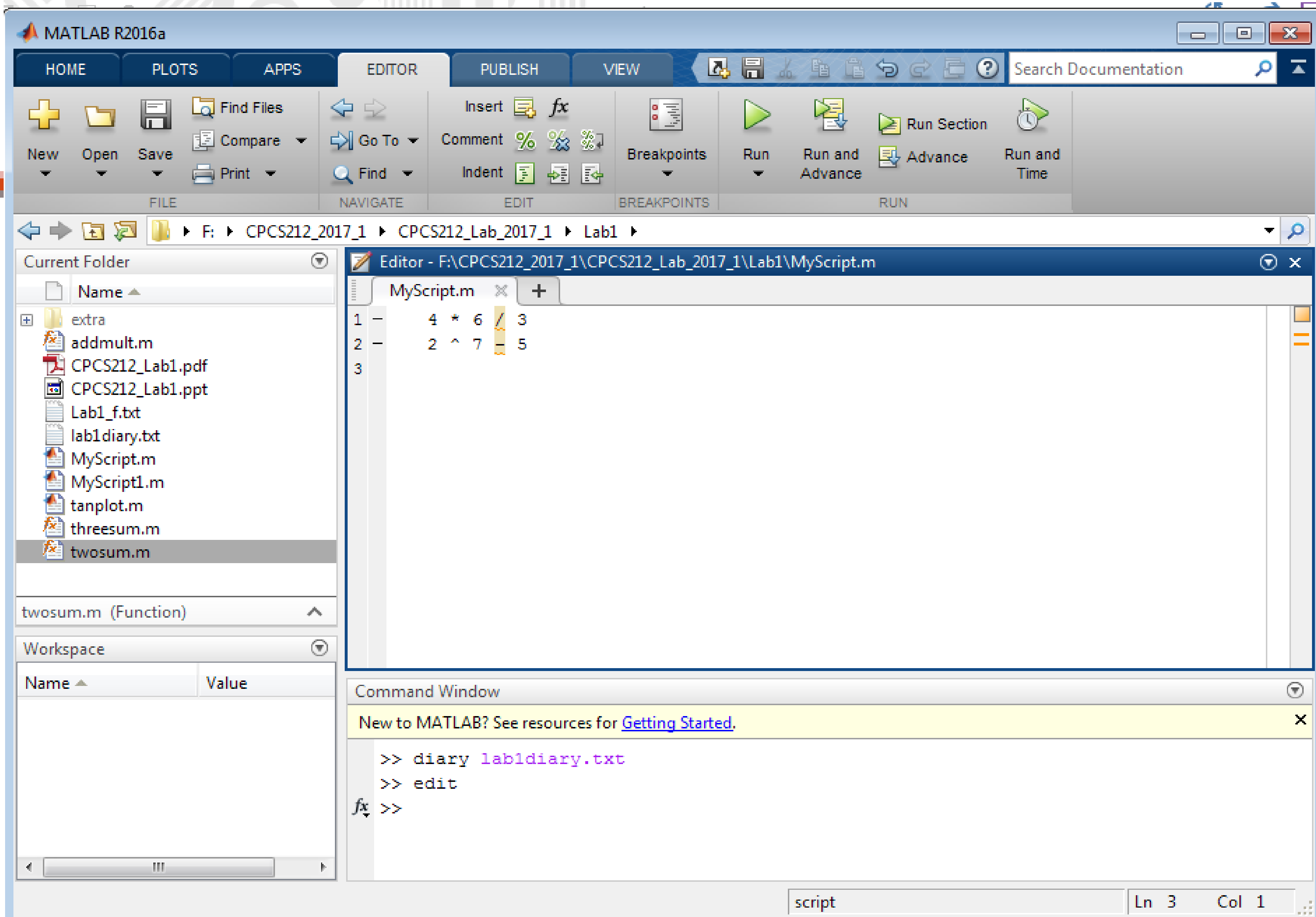
  for example: nameoffile.m

Keep in mind when using script files:

- The name of a script file MUST begin with a letter, and may include digits and the underscore character, up to 31 characters.

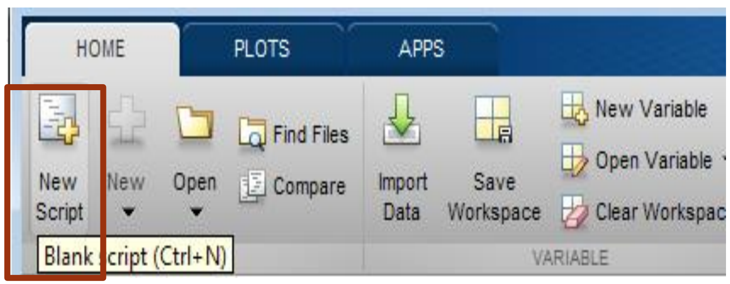
- Do not give a script file the same name as a variable.

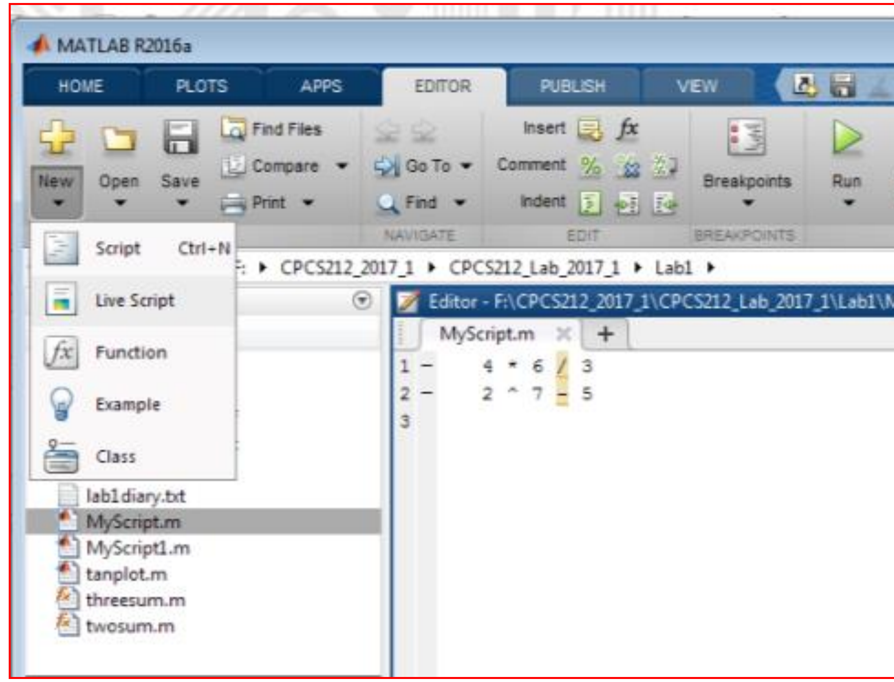
- Do not give a script file the same name as a MATLAB command or function. You can check to see if a command, function or file name already exists by using the ***exist*** command.


MATLAB R2016a
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
Search Documentation
New
Open
Save
Find Files
Compare
Print
FILE
Go To
Find
NAVIGATE
Insert
Comment
Indent
EDIT
Breakpoints
BREAKPOINTS
Run
Run and Advance
Run Section
Advance
Run and Time
RUN
F:
CPCS212_2017_1
CPCS212_Lab_2017_1
Lab1
Current Folder
Name
extra
addmult.m
CPCS212_Lab1.pdf
CPCS212_Lab1.ppt
Lab1_f.txt
lab1diary.txt
MyScript.m
MyScript1.m
tanplot.m
threesum.m
twosum.m
twosum.m (Function)
Workspace
Name
Value
Editor - F:\CPCS212_2017_1\CPCS212_Lab_2017_1\Lab1\MyScript.m
MyScript.m
1 – 4 * 6 / 3
2 – 2 ^ 7 - 5
3
Command Window
New to MATLAB? See resources for Getting Started.
>> diary lab1diary.txt
>> edit
>>
script
Ln 3
Col 1

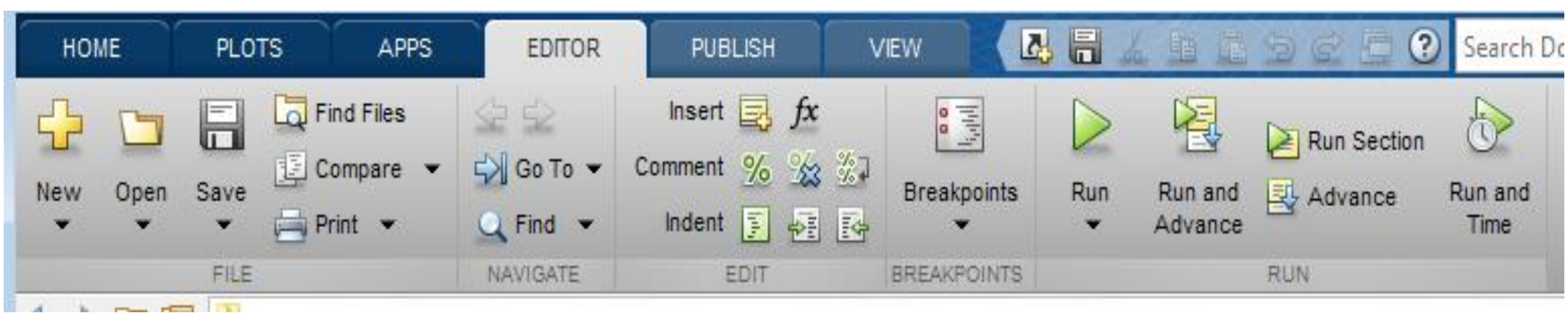
MATLAB Editor (Script files)

- Opening MATLAB Editor



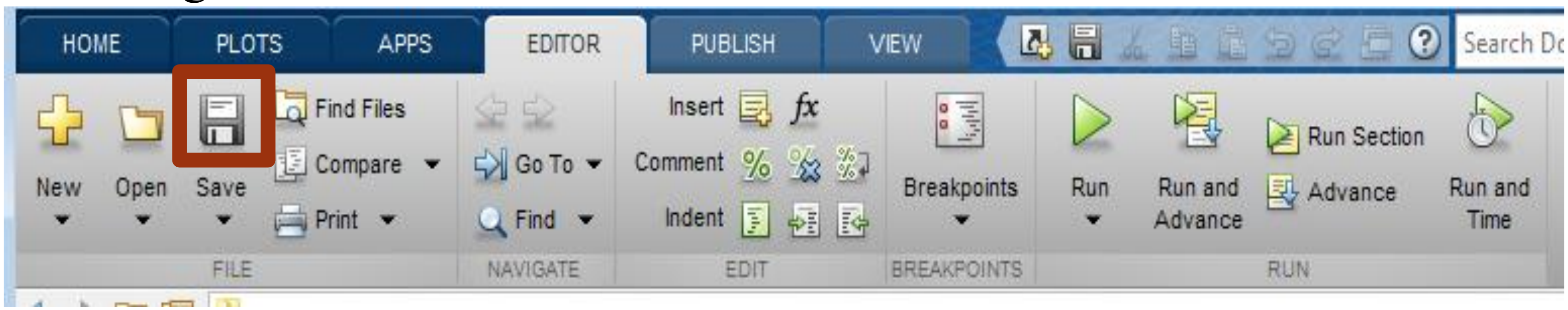


- The MATLAB editor navigation



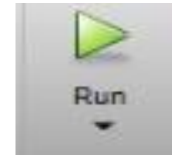
MATLAB Editor (Script files)

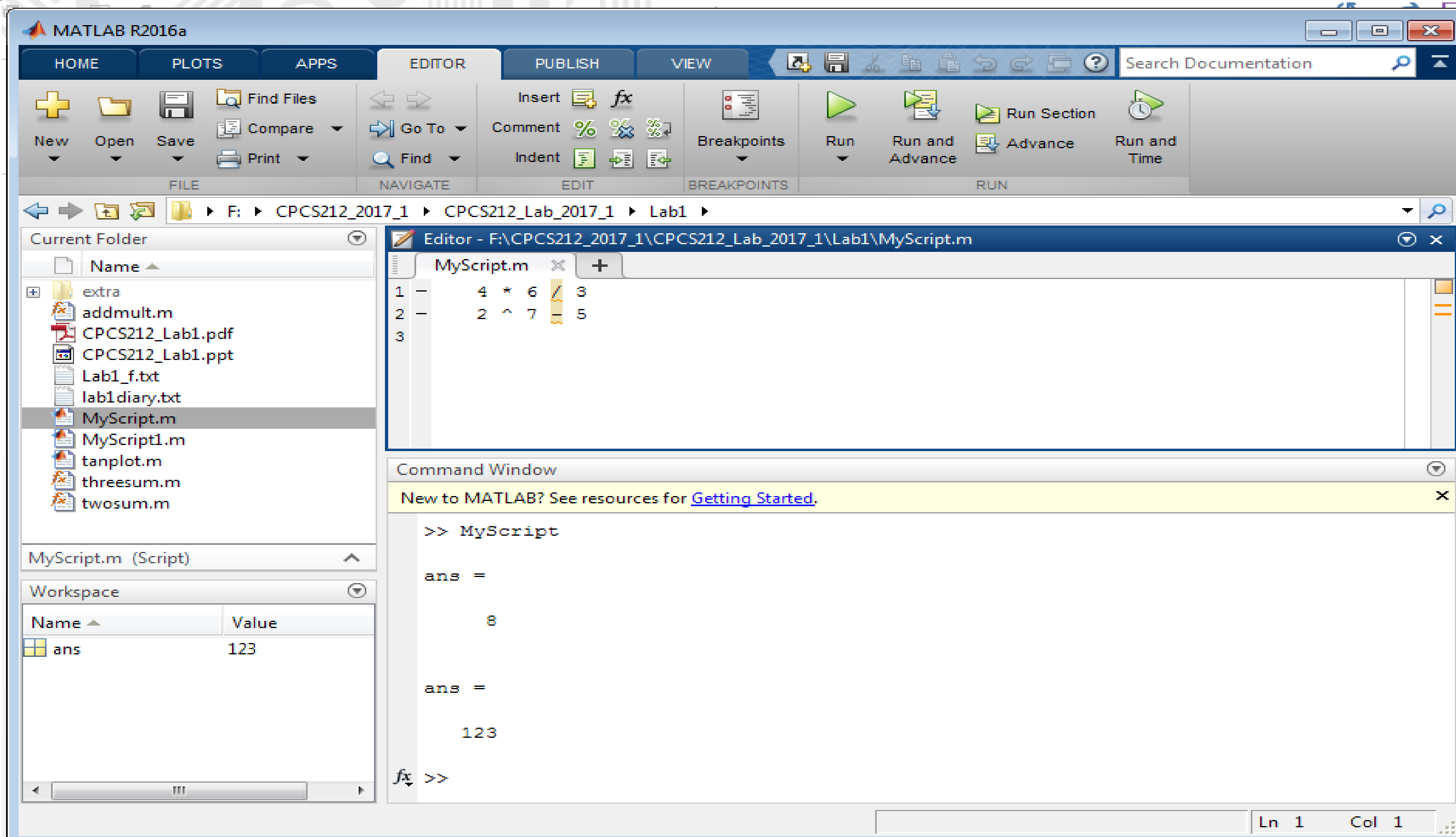
- Saving



- Running your script

Running your script





Getting Help

- **From the Command Window (help , lookfor , doc)**
- **Help Browser Window**
- **Function Browser (SHIFT+F1)**
- **Examples / Demos**

Getting Help

Getting Help	Description
Help Browser Window	The Help Browser enables you to search and view documentation for MATLAB and other Math Works products. Help button in resources tab click Documentation
Function Browser	If you don't know the name of a function, you can press **SHIFT+F1** at the command window to open a simple keyword search interface. Type a word and MATLAB will give a list of matching commands.
From the Command Window	After you type in the name of a MATLAB command followed by an open bracket, MATLAB will provide syntax hints for the command or press **Ctrl+F1** .

Help Functions

- help funcname: Displays in the Command window a description of the specified function funcname.

- lookfor topic: Displays in the Command window a brief description for all functions whose description includes the specified key word topic.

- doc funcname: Opens the Help Browser to the reference page for the specified function funcname, providing a description, additional remarks, and examples.

Lab Exercise

Command	Exercise	Output
help	help sine	??
	help sin	??
lookfor	lookfor sine	??
	lookfor sin	??
doc	doc sine	??
	doc sin	??

Variable Names

- Variables are automatically allocated
- MATLAB does not require declaration of variables
- Rules for declaring a variables in MATLAB:
  a. Variable names must begin with a letter
  b. The rest of name can contain letters, digits and underscore characters
  c. Variables names must contain less than 32 characters
  d. MATLAB is case-sensitive
  e. Variable should not be the name of a built-in variable , built-in function.

Built-in MATLAB Variables (1)

Name	Meaning
`ans`	value of an expression when that expression is not assigned to a variable
`eps`	floating point precision
`pi`	π, $(3.141492\ldots)$
`realmax`	largest positive floating point number
`realmin`	smallest positive floating point number
`Inf`	∞, a number larger than `realmax`, the result of evaluating $1/0$.
`NaN`	not a number, the result of evaluating $0/0$

Built-in MATLAB Variables (2)

Rule: Only use built-in variables on the right hand side of an expression. Reassigning the value of a built-in variable can create problems with built-in functions.

Exception: `i` and `j` are preassigned to $\sqrt{-1}$. One or both of `i` or `j` are often reassigned as loop indices. More on this later.

Format Types

Type	Result
short	Scaled fixed point format, with 4 digits after the decimal point. For example, 3.1416
long	Scaled fixed point format with 14 to 15 digits after the decimal point for double; and 7 digits after the decimal point for single. For example, 3.141592653589793
short e	Floating point format, with 4 digits after the decimal point. For example, 3.1416e+000
long e	Floating point format, with 14 to 15 digits after the decimal point for double; and 7 digits after the decimal point for single. For example, 3.141592653589793e+000
short g	Best of fixed or floating point, with 4 digits after the decimal point. For example, 3.1416
long g	Best of fixed or floating point, with 14 to 15 digits after the decimal point for double; and 7 digits after the decimal point for single. For example, 3.14159265358979
short eng	Engineering format that has 4 digits after the decimal point, and a power that is a multiple of three. For example, 3.1416e+000
long eng	Engineering format that has exactly 16 significant digits and a power that is a multiple of three. For example, 3.14159265358979e+000

Exercise 1

1-(a)

What are the values of the reserved variables pi, eps, realmax, and realmin?

```
>> % Exercise 1:

>> % 1 – (a)
>> pi
>> eps
>> realmax
>> realmin
```

1-(b)

Use the "***format long***" command to display ***pi*** in full precision and "***format short***" to return Matlab to its default, short, display.

```
>> % 1 – (b)
>> format long
>> pi
>> format short
```

1-(c)

Choose a value and set the variable *x* to that value.

```
>> % 1 – (c)
>> x = 2
```

1-(d)

What is the square of *x*? Its cube?

```
>> % 1 – (d)
>> x^2
>> x^3
```